Tony Chen

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EDUCATION

University of California, Los Angeles Expected 2030
B.S., Electrical Engineering

Shady Side Academy, Pittsburgh, PA 2022 – 2026
- GPA 3.96/4.00. National Merit Finalist (Pennsylvania). ACT 36 composite; SAT 1560.
- AP Calculus BC (5), AP Computer Science A (5), AP Physics C: Mechanics (4), AP Physics C: E&M (4); coursework in linear algebra, differential equations, multivariable calculus, and calculus-based physics.

EXPERIENCE

NASA, Embedded Systems Engineering Intern (paid) Summer 2025
- Developed an autonomous mapping and placement pipeline in ROS2 on a rover platform.
- Built a LIDAR detection system to map an unknown area and generate a spatial model.
- Applied OpenCV and AprilTag detection to identify and place areas within the newly mapped space.

TECHNICAL SKILLS

Programming: Python, Java, C++, ROS2, OpenCV
CAD / CAM: Autodesk Fusion 360 for full-assembly modeling, part design, and CAM toolpaths
Fabrication: FDM 3D printing (plastic parts), CNC milling (metallic parts), mechanical assembly
Electronics: Motor drive, electronic speed controllers, stepper motor control, LiPo power systems

ENGINEERING PROJECTS

Automated Tracking Telescope 2025 – 2026
- Locates and tracks deep-sky objects using two stepper motors on independent axes.
- Camera attachment for long-exposure astrophotography, plus swappable eyepieces for variable magnification.
- Shutter system for brightness control; target magnification 200×, enough to resolve Jovian bands and Saturn's rings.

Custom Remote-Controlled Race Car 2024 – 2025
- 3D-printed chassis with a dual-motor drivetrain, independent electronic speed controllers, and a shared 5200 mAh 11.1 V LiPo pack.
- Ackermann steering geometry so the inner and outer wheels track at different optimal angles, tightening the turning radius and keeping the tires from scrubbing in corners.
- 15° caster angle to generate self-aligning torque, improving straight-line stability at speed and steering return after turns.

Science Olympiad Electric Vehicle 2024 – 2025
- Arduino-controlled vehicle that runs a track as fast as possible and stops at a target point, scored on combined time and stopping distance. The 7–10 m target distance is announced only after impound, so the vehicle had to be recalibrated to an unknown distance and run twice inside an 8-minute window.

LEADERSHIP AND ACTIVITIES

Leader, Robotics Club and FTC Team at Shady Side Academy 2024 – 2026
- Sole club leader: recruited, organized the team, led fundraising, and directed design, build, programming, and competition.
- Modeled the complete robot chassis from scratch in Fusion 360 using goBILDA components, adding a formal CAD phase the team had never had before, and wrote the control code in Java.
- Drove recruiting to ~80 signups from 600 students; raised ~$600 via crowdfunding and a bake sale to close the equipment funding gap.

Computer Science Departmental Associate at Shady Side Academy 2024 – 2026
- One of five students selected to represent the CS department at school events, tutor peers, and promote computer science programs.

Writing Center Scribe at Shady Side Academy 2025 – 2026
- Admitted as 1 of 20 applicants to the Theory and Process of Writing course and selected to serve as a writing center scribe.

Science Olympiad 2024 – 2026
- Competed in Electric Vehicle, Towers, Bungee, and miscellaneous study events; consistent medal placements across all events.

Music and Athletics 2014 – 2026
- First violin, Pittsburgh Youth Philharmonic Orchestra (2019–2026); ABRSM Piano Grade 8 and Theory Grade 5, both with Distinction; JV soccer, basketball, and track & field.